

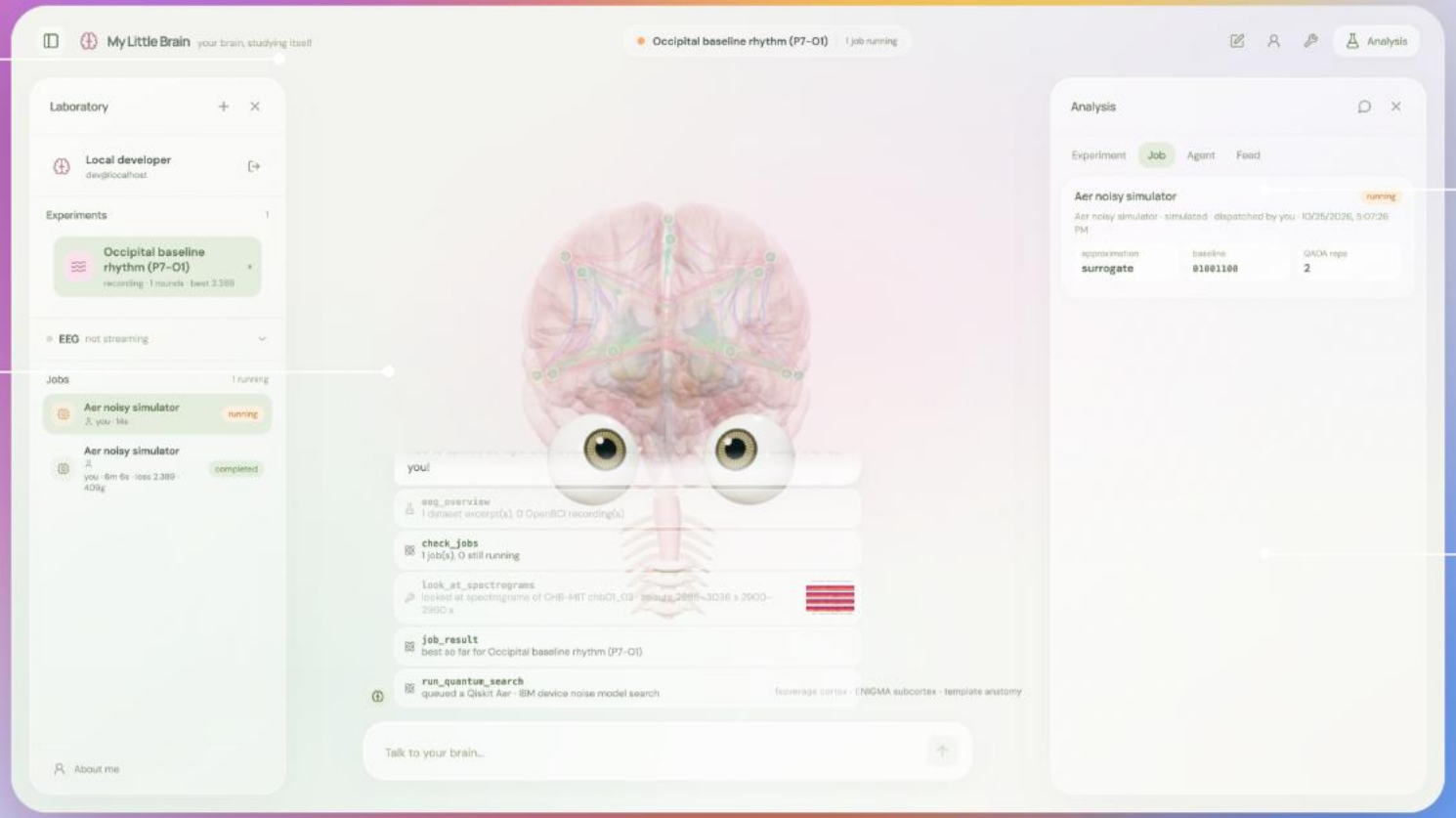
# My Little Brain

# It knows you

your notes · your live EEG · your questions

it watches your live EEG

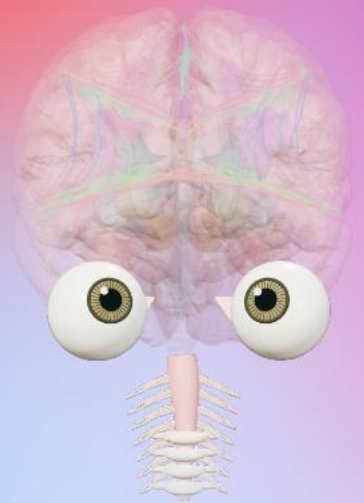
you chat with your brain




it designs quantum circuits

results land here

# It studies itself



IONQ
IonQ · trapped ions

IBM
IBM · heavy-hex

qBraid
qBraid · Rigetti/IQM/AQT

Extropic · THRML

Gemini · vision + agent

OpenBCI · your EEG

**48 / 256**
simulations to recover a hidden brain state

**42 gates**
a circuit Gemini wrote itself (from 106)

**52 vs 454**
gates on IonQ vs IBM, same problem

**τ 16→55 ms**
seizure windows point to slower synapses

OpenQASM written by Gemini 4 Argon · Qiskit Aer, IBM Torino noise · job f8fdb905…

```
OPENQASM 2.0;
…
qreg q[8];
creg meas[8];
ry(1.7) q[7];
x q[6];
x q[5];
x q[2];
ry(2.5) q[1];
cx q[7],q[6];
cx q[6],q[4];
cx q[7],q[3];
ry(-0.55) q[3];
…
```

model hypotheses, not diagnoses · no quantum-advantage claim

# EEG → quantum state

## 2,000 voltages a second → 10 numbers

8 channels × 250 Hz, condensed into band powers

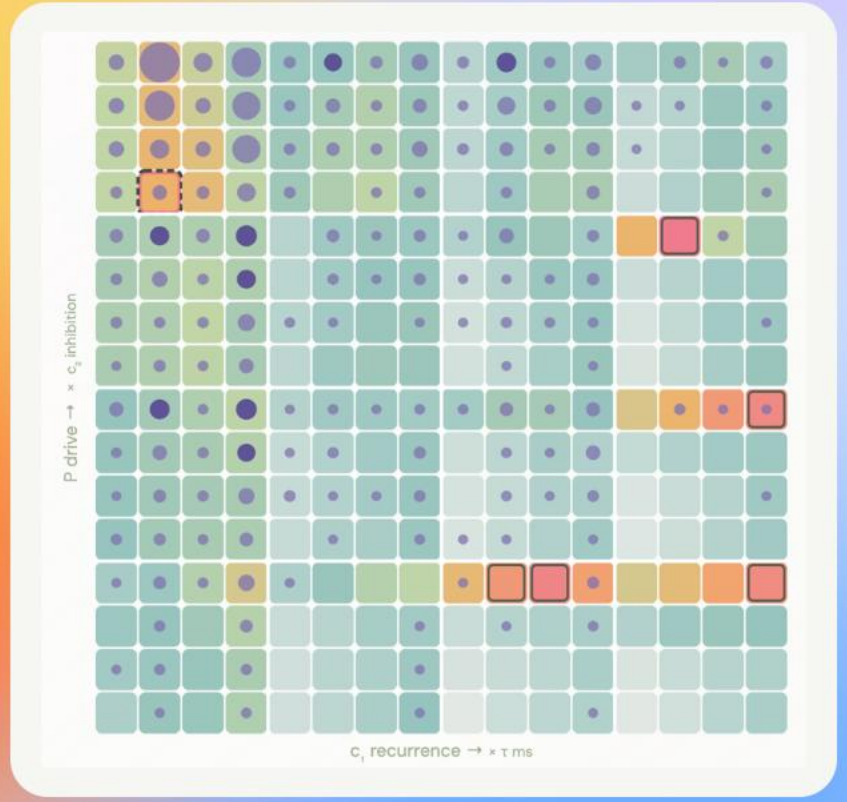
## 8 qubits hold all 256 possible brains

4 model dials × 2 qubits · a $2^8$-dim Hilbert space $|\psi\rangle = \Sigma\ \alpha_i\ |i\rangle$

## Your EEG decides which brains get louder

the mismatch becomes the circuit's energy; QAOA amplifies the best fits

## Measure → re-simulate → verify



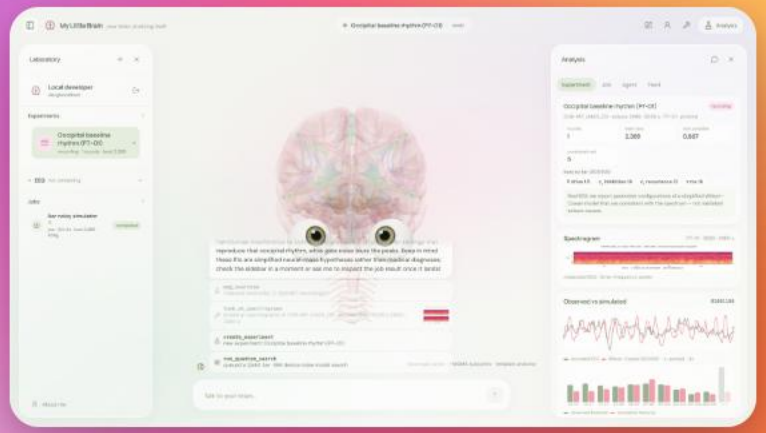

all 256 brains, scored against your EEG (from the app)

we encode hypotheses, not raw EEG · each extra bit per dial multiplies the space · no quantum-advantage claim at 256

# Behind the scenes



**OpenBCI Ultracortex**
Web Serial · Chrome



**The app**
chat · 3D brain · experiments

**Gemini 4 Argon**
agent · Cloud Run

GOOGLE CLOUD

Cloud Run · Firestore · Storage · Cloud Tasks

Secrets · Registry · Sign-In

QAOA · thermodynamic sampling

IONQ · IBM · qBraid · Qiskit Aer · Extropic THRML

validated by forward simulation

4 brain parameters × 2 bits = 8 qubits = 256 hypotheses · nonlinear loss → QUBO surrogate → exact Ising map

# The team



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quantum applications · quantum ML



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